

StudyEezy Full PRD

StudyEezy - Full Product Requirements Document (PRD)

Version 1.0 | Mobile-First Responsive Web Application

StudyEezy is an AI-powered collaborative learning and assessment platform that helps students study actively, test knowledge, receive AI-reviewed feedback, and continue learning offline.

Product Vision

Help students learn actively rather than passively through collaboration, assessments, revision tools, and AI-powered feedback.

Problem Statement

Many students rely on instant-answer AI tools, leading to passive learning and poor retention. Students need structured revision, peer learning, meaningful feedback, and offline accessibility.

Target Audience

Primary: University students.

Secondary: WAEC, NECO, and JAMB candidates.

Tertiary: Self-learners and certification candidates.

Core Features

Authentication, Study Material Uploads, OCR, Voice Input, AI Summaries, Flashcards, Quiz Generation, AI Chat, Study Rooms, AI Review Mode, Offline Learning.

Authentication

Sign Up, Login, Logout, Email Verification, Password Reset. Email verification required before protected features can be accessed.

Study Material Management

Users can upload PDF, DOC/DOCX, images, and camera captures. OCR converts images and captured notes into searchable AI-processable text.

Voice Input

Speech-to-text supported in AI Chat, Study Rooms, Quiz Answers, and Search.

AI Summary Generation

Users explicitly choose one summary type: Short Summary, Detailed Summary, Revision Notes, Key Concepts, or Exam Prep Summary. AI must not automatically choose summary type.

Flashcards

Generated from uploaded materials and summaries. Flashcards remain accessible offline and can be marked as mastered.

Quiz Generation

Users configure question type (MCQ, Short Answer, Theory, Mixed), question count, and difficulty before generation.

Quiz Rules

No hints, no grading, and no answer reveals until final submission. AI evaluates only after the quiz is submitted.

Quiz Evaluation

Display score, grade, correct answers, incorrect answers, explanations, weak-topic analysis, recommended topics, and retake-failed-questions option.

AI Chat

Used for concept clarification, explanations, examples, and learning support. Requires internet connection.

Study Rooms

Users create rooms, invite members via links, and collaborate through questions and answers.

Study Room Roles

Owner: create/delete room, invite/remove members, ask questions, answer questions, request AI reviews, view room history.

Member: ask questions, answer questions, request AI reviews, view room history.

AI Review Mode (Signature Feature)

Question → Peer Answer → AI Review Request → AI Evaluation. AI does not answer automatically. AI evaluates only when requested.

AI Review Output

Score, strengths, missing concepts, suggested improvements, and model answer.

Offline Experience

Available Offline: summaries, flashcards, uploaded materials already synced, quiz history, room history.

Online Only: AI chat, AI reviews, OCR processing, summary generation, quiz generation, live room messaging.

Technology Stack

Frontend: Next.js 15, TypeScript, Tailwind CSS, shadcn/ui.

Backend: Next.js Server Actions.

Database: Firebase Firestore.

Authentication: Firebase Authentication.

Storage: Firebase Storage.

Realtime: Firestore Realtime Listeners.

AI: DeepSeek API.

OCR: Google Vision API.

Voice: Web Speech API.
Offline: PWA, IndexedDB, Firestore Offline Persistence.
Deployment: Vercel.

Success Metrics

Quizzes completed, flashcards reviewed, summaries generated, study rooms created, AI reviews requested, DAU, WAU, retention, and offline usage.

Non-Goals (MVP)

No video calls, audio calls, whiteboards, teacher accounts, LMS functionality, marketplace, subscriptions, premium plans, social feed, native mobile apps.